

AcePharm — Web Platform Brief

Consolidate the web architecture, question engine, and clinical progress platform (Next.js + Prisma/Postgres)

Prepared for	Haseeb Sajjad / Lead Engineering Team
Client & Product	Takween Centre UK Ltd · AcePharm Web Platform (acepharm.co.uk)
Project type	Production Build — Milestone 1 to 7 Implementation & Foundation Delivery
Reference documents	AcePharm Developer Brief v2.1, Website Copy v2.0, Interactive Prototype
Core Stack	Next.js (App Router, Strict TS) · Tailwind CSS & shadcn/ui · PostgreSQL + pgvector (Prisma ORM) · Auth.js (Credentials + Magic Link) · Stripe Billing · Anthropic Claude (Messages API) · Resend · PostHog · Sentry · Vercel
Core feature set	Public Marketing Site & SEO · Multi-step Onboarding · High-frequency Question Engine (Learn/Timed) · Dual Attempt Storage · Multi-metric Progress Analytics · Grounded AI Tutor (Ace) · Content Authoring Admin
Distribution (QA)	Staging preview deployment with seeded clinical question bank and Stripe test mode ahead of production promotion

Purpose

AcePharm is the dedicated, high-yield digital revision platform tailored specifically for UK Master of Pharmacy (MPharm) undergraduates (Years 2 to 4) and trainee pharmacists preparing for the General Pharmaceutical Council (GPhC) registration assessment. Unlike generic exam preparation tools, AcePharm bridges the structural clinical gap through a tightly coupled core learning loop: **Practise** → **Understand** → **Analyse** → **Improve**.

This build consolidates the entire web infrastructure into seven structured milestones. It establishes strict schema isolation, dual-store attempt logging (preserving pristine first-attempt benchmarks permanently alongside active practice data), sub-300ms question interactions, and Ace: a contextually retrieved, strictly grounded AI tutor operating exclusively on peer-reviewed AcePharm clinical assets. This is production-grade clinical software where data integrity, clinical authoring audit trails, and strict role permissions are paramount.

Design & UI — Extend the Existing System, Don't Reinvent It

The interactive HTML prototype and Design System tokens serve as the absolute source of truth for all visual hierarchy, element sizing, layout spacing, and micro-interactions. The engineering team must directly consume the formal design tokens rather than approximating styling:

- Color Palette:** Primary Ace Indigo (#4F46E5), Deep Indigo (#3730A3), Ink text (#111827), Slate secondary text (#64748B), Canvas background (#F8F AFC), Surface card (FFFFFF), Border line (#E2E8F0). Supporting semantic tokens: Clinical Teal (#0F766E), Success (#15803D), Warning (#B45309), Danger (#B91C1C).
- Typography & Layout:** Geist Sans for clean user interfaces and editorial body text; Geist Mono for clinical drug calculations, formulas, question public IDs, and keyboard shortcuts. Stems and explanation copy are rendered at 18px with a comfortable 65–75 character measure.
- Touch & Micro-interactions:** Strict 8-point spacing grid, 10–12px border radii, subtle clean borders instead of heavy box shadows, and a non-negotiable 44×44px minimum interactive touch target.

What to Build

1. Authentication, Onboarding & RBAC Foundation (Milestone 1)

- Implement native Auth.js credentials login and email magic-link flows without third-party auth vendors. Server-side session validation on every protected route.
- Strict Role-Based Access Control (Admin, Clinical Author, Reviewer, Learner) gating both frontend routing and Next.js API endpoints. Unauthorized requests must return clean HTTP 403 Forbidden states rather than generic errors.
- Implement the complete 5-step onboarding flow from Website Copy (Stage, Primary Goal, Assessment Date, Daily Question Target, School of Pharmacy) with full skip capability.

2. Database Schema & Dual-Store Attempt Logging (Milestone 1 & 4)

- Deploy PostgreSQL with Prisma ORM and pgvector extension across a dedicated UK/EU region database instance.

- **Dual Attempt Storage:** Store the pristine first attempt in `QuestionFirstAttempt` (immutable, never modified or wiped) while ongoing and repeat practice records write to `QuestionAttempt`. Category resets wipe practice records while leaving first-attempt baseline records permanently intact.
- Store financial amounts strictly in integer pence. Enforce explicit foreign key cascade deletion policies, database-level soft deletes, and composite indices across user progress and question lookup tables.

3. Content Authoring, Question Review & CSV Importer (Milestone 2)

- Curriculum taxonomy management covering BNF-aligned body systems, high-risk drug classes, OTC scenarios, and pharmaceutical calculations.
- Clinical Question Editor enforcing mandatory metadata: stem, 5 distinct options, individual rationale for every option, subtopic notes, author attribution ('AcePharm Clinical Editorial Team'), and full version audit history.
- Robust CSV/spreadsheet importer tool with dynamic column mapping, syntax validation, duplicate detection, and actionable pre-import error reporting. Validate and import Seed Question Bank v0.1 (135 clinical questions).

4. High-Performance Question Practice Engine (Milestone 3)

- **Session Builder:** Dynamic filtering by category, high-risk flags, difficulty, past attempt history (All, Unused, Incorrect, Flagged), and mode (Learn vs. Timed).
- **Strict Answering Rules:** Confidence rating (Low, Medium, High) must be captured before submission. Absolutely no colour signals (green/red) prior to submission. After submission, render clear accessible SVG icons and text labels alongside colour. Disallow automated advancing.
- **Clinical Feedback & Tools:** Full explanation accordion with per-option rationales, subtopic revision notes, hide-options mode, flagging, personal notes, and instant report-a-question modals with admin routing.

5. Progress Analytics & Multi-Metric Dashboard (Milestone 4)

- Display five uncollapsed metrics: First-Attempt Accuracy, Practice Accuracy, Repeat Accuracy, Confidence Calibration, and Curriculum Coverage.
- Visual Topic Map using the formal 6-state progression model: *Not Started* → *First Pass* → *Needs Attention* → *Developing* → *Secure* → *Due for Review*.
- Defensible recommendation engine surfacing actionable next sessions with clear contextual reasons (e.g. '3 high-risk cardiovascular questions due for review'). Meaningful streaks rewarding completed sessions over simple logins.

6. Grounded AI Tutor Layer — 'Ace' (Milestone 5)

- Semantic retrieval engine indexing explanation rationales, subtopic clinical notes, and BNF reference summaries using pgvector embeddings.
- Strictly grounded RAG prompt architecture: Claude generates responses solely from retrieved AcePharm content with verifiable citations. When content is absent, Ace explicitly declines to answer rather than hallucinating facts. Real-patient queries are strictly refused.
- Contextual question panel, highlight-to-ask inline interactions, calculation step coach, and spaced-repetition flashcard generator with fair-use throttling.

7. Public Marketing, Stripe Billing & Launch Hardening (Milestone 6 & 7)

- Next.js App Router public marketing site, editorial standards showcase, interactive question demo, legal hub, and blog with administrative CMS.
- Full Stripe Checkout and Customer Portal integration supporting Explorer (free 30 questions), Monthly (£16/mo), and Annual (£120/yr) tiers. Enforce webhook signature validation and frictionless online cancellation.
- Security hardening: Rate-limiting on auth and AI endpoints, CSP headers, automated daily database backup restoration verification, and WCAG 2.2 AA accessibility audit.

Technical Requirements

- **Stack Compliance:** Next.js 14+ (App Router), Strict TypeScript, Tailwind CSS, shadcn/ui, PostgreSQL with pgvector, Prisma ORM, Auth.js, Stripe, Anthropic Claude API, Resend, Sentry, and PostHog (EU).
- **Performance Budgets:** Marketing LCP under 2.0s on 4G, INP under 200ms, CLS under 0.1. Perceived answer submission feedback under 300ms with optimistic UI reconciliation. Instant prefetching between questions.
- **Security & Privacy:** Passwords hashed via Argon2/bcrypt (cost 12+). Strict server-side authorization on all routes. No API secrets or environment variables committed to Git. Non-essential cookies blocked prior to explicit user consent.

- **Accessibility Standard:** Strict WCAG 2.2 AA compliance across all public and learner surfaces, including full keyboard navigation, visible focus rings, ARIA live regions for answer reveals, and colorblind-safe visual feedback.

What to Submit

Upon completion of each project milestone, submit a comprehensive Milestone Verification Document containing the following mandatory fields:

#	Field	What to Include
1	Staging Access	Live Vercel staging URL, plus working test credentials for Admin, Clinical Author, and Learner accounts.
2	Scope & Milestones Covered	Specific milestones and sub-features delivered in this submission, explicitly noting any deferred items.
3	Database & Schema Integrity	List of executed Prisma migrations, confirmation of pgvector indexing, and dual-store attempt logging verification.
4	GitHub Repository	Full repository URL, branch reference, and a clean commit log reflecting logical, incremental progress.
5	Loom Screen Walkthrough	5–8 minute video demonstrating: onboarding, session builder, question practice loop with confidence capture, full explanation layout, Ace contextual RAG citations, topic map analytics, and Stripe checkout/cancellation.
6	What Was Built	Concise 5 to 10 line summary of user-facing changes and core engineering outcomes delivered.
7	Challenges & Resolutions	Specific technical challenges encountered (e.g. pgvector query optimization, Auth.js edge routing) and exact architectural fixes applied.
8	Verification & Test Script	Step-by-step instructions for reproducing end-to-end testing, including seed data verification and Stripe CLI webhook simulations.
9	Known Limitations	Edge-case behaviours or secondary polish items planned for future milestones.
10	Path to Production Plan	Written multi-paragraph plan addressing production migration, pgvector index warming, disaster recovery, and automated backup restoration.
11	Submission Date	Official calendar date of milestone delivery.

QA Review Criteria

#	Assessment Point
1	Is first-attempt performance permanently separated from repeat practice, with baseline data surviving category resets?
2	Is confidence captured strictly prior to submission, with zero color indication before submission and multi-modal feedback afterwards?
3	Does every single question option provide an independent, clinically sound rationale attributed to the AcePharm Clinical Editorial Team?
4	Does Ace answer strictly and solely from retrieved AcePharm content with explicit citations, declining non-covered or real-patient prompts?
5	Are all admin and API endpoints protected by server-side role enforcement, returning HTTP 403 upon unauthorized access?
6	Does the platform meet performance thresholds (<300ms answer feedback, prefetching) and adhere to WCAG 2.2 AA accessibility?
7	Are Stripe subscriptions and online cancellation workflows functioning seamlessly without manual admin intervention?
8	Has the entire seed question bank (135 questions) been successfully ingested and verified through the content importer?

Path to Production

Prior to cutting the production release, the engineering team must satisfy a rigorous four-pillar readiness checklist. First, database integrity must be formally verified on a staging replica: all Prisma migrations must execute idempotently with proven rollback scripts, composite indexes must cover user question history, and automated backup restoration must be completed successfully without data loss.

Second, the AI grounding pipeline must undergo regression testing across an automated evaluation suite of 50+ benchmark clinical queries. This guarantees that similarity thresholds in pgvector prevent hallucinations, that citations point to active subtopic notes, and that out-of-domain and patient-specific prompts are safely deflected.

Third, the API contract powering the web application must be frozen, documented, and versioned. Because the forthcoming AcePharm Flutter mobile application will consume these exact endpoints for authentication, practice sessions, and progress metrics, no breaking schema changes may occur without backwards-compatible API versioning.

Finally, all legal, privacy, and compliance obligations must be satisfied: cookie consent banners must strictly gate non-essential trackers, GDPR-compliant account deletion and data export endpoints must be functional, and Stripe production webhooks must be verified under end-to-end live testing.